

Ni-Catalyzed Enantioselective Suzuki–Miyaura-Type Coupling of Styrene Oxides with Arylboronic Acids

Xiao-Yu Lu,* Hao-Yu Dong, Jun-Long Zhang, Jin-Tao Tao, Si-Nan Liu, and Yi-Mei Wu

Cite This: *ACS Catal.* 2026, 16, 2417–2426

Read Online

ACCESS |



Metrics & More



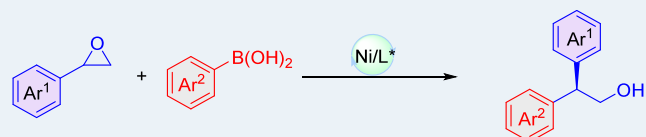
Article Recommendations



Supporting Information

ABSTRACT: Organoboron compounds are universally recognized as highly valuable synthetic building blocks, while the Suzuki–Miyaura cross-coupling has firmly established itself as a reliable platform for C–C bond formation. These salient features have fostered its widespread application across medicinal chemistry and materials science. Herein, a nickel-catalyzed, exogenous base-free, enantioselective Suzuki–Miyaura-type coupling of styrene oxides with arylboronic acids is reported. This reaction demonstrates high regioselectivity and enantioselectivity, a broad substrate scope, versatility for the late-stage modifications of bioactive molecules, and capability for synthesizing chiral drug precursors. Additionally, aryl aziridines can be employed as alternative coupling partners. Mechanistic studies confirm a stereoconvergent pathway involving radical intermediates, with in situ-generated alkoxide anions acting as the essential base. This methodology provides a practical approach for synthesizing enantioenriched diaryl alcohols and amines, which are valuable structural skeletons in pharmaceuticals and natural products.

KEYWORDS: nickel catalysis, arylboronic acids, enantioselective coupling, styrene oxides, Suzuki–Miyaura



- Good functional group tolerance
- Exclusive regioselectivity
- Without exogenous base
- Up to 94% ee

INTRODUCTION

Organoboron compounds stand as highly important synthetic building blocks, and the Suzuki–Miyaura cross-coupling has firmly established itself as a robust methodology for C–C bond formation between halides/pseudohalides and boronic acid-containing substrates.¹ Compared to other transition-metal-catalyzed cross-coupling systems, it offers distinct advantages, including the broad availability of boronic acids, straightforward reagent handling, and mild reaction conditions. These merits have driven its widespread application in the pharmaceutical industry, underpinning C–C bond construction in numerous medicinal agents.² Separately, epoxides are recognized as privileged building blocks in organic synthesis owing to their ready accessibility from olefins or ketones, as well as their unique intrinsic ring strain and versatile reactivity.³ These properties facilitate diverse transformations into high-value alcohols and complex molecular architectures (Scheme 1a). Among the various reactions involving epoxides, asymmetric ring-opening with diverse nucleophiles emerges as a practical and attractive strategy for constructing chiral alcohols—key intermediates in the synthesis of pharmaceuticals, natural products, and agrochemicals.⁴ Despite the maturity of this field, asymmetric ring-opening strategies often rely on stereospecific ring scission, either through kinetic resolution of racemic epoxides or desymmetrization of meso-epoxides, typically employing heteroatom-centered or soft nucleophiles.^{4,5} While asymmetric catalytic C–C bond formation with epoxides can also be achieved using organo-

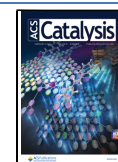
lithium or organomagnesium reagents,⁶ these methodologies suffer from harsh reaction conditions and limited functional group tolerance. Thus, the development of chiral catalyst-controlled, stereoconvergent C–C bond-forming reactions for racemic epoxides remains highly valuable for both synthetic utility and fundamental research.⁷ In recent years, transition-metal-catalyzed asymmetric cross-electrophile ring-opening reactions involving aryl electrophiles have emerged as a powerful platform for accessing chiral diaryl or aryl-substituted alcohols. Notably, in 2015, Zhao and Weix reported the first nickel/titanocene cocatalytic system for the enantioselective reductive cross-coupling of meso-epoxides with aryl halides,⁸ where chiral titanocene dictated stereoselectivity through epoxides activation. Meanwhile, the Yamamoto group described a nickel-catalyzed arylation of 3,4-epoxyalcohols enabled by chiral ligands,⁹ delivering cross-coupled products with excellent enantio- and diastereoselectivity. In 2021, Doyle and co-workers further advanced this field with a photoinduced nickel-catalyzed reductive coupling of epoxides and (hetero)-aryl iodides,¹⁰ which afforded enantioenriched 2,2-diaryl alcohols via radical intermediates (Scheme 1b).

Received: October 23, 2025

Revised: January 6, 2026

Accepted: January 8, 2026

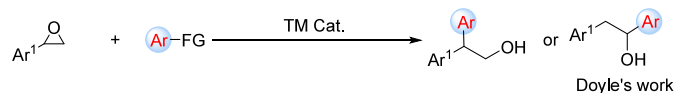
Published: January 16, 2026



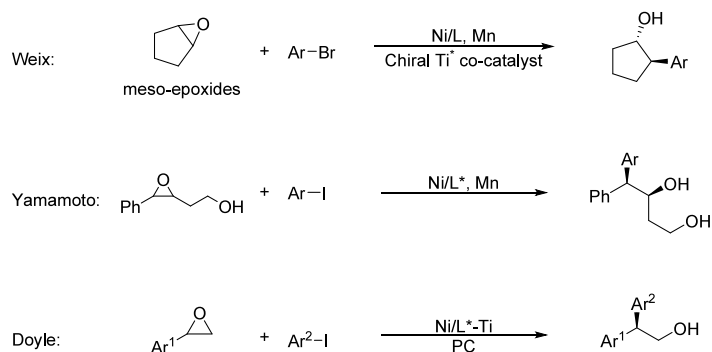
Scheme 1. Enantioselective Ring-Opening Cross-Couplings of Epoxides: (a) Transition-Metal Catalyzed Coupling of Epoxides; (b) Asymmetric Reductive-Coupling of Epoxides; (c) Enantioconvergent Coupling of Spiroepoxyoxindoles with Alkenylboronic Acids; (d) This Work: Asymmetric Suzuki-Type-Coupling of Epoxides with Arylboronic Acids; (e) Representative Bioactive Compounds Containing Chiral 1,1-Diaryl Alkane

Previous work:

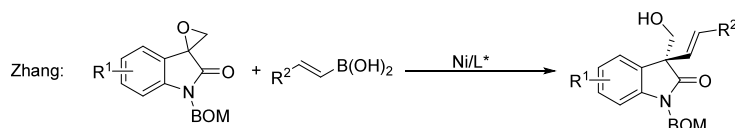
a. Transition-metal catalyzed coupling of epoxides



b. Transition-metalcatalyzed asymmetric reductive-coupling of epoxides with aryl electrophilic

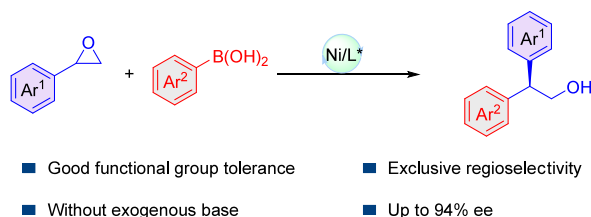


c. Ni-catalyzed enantioconvergent coupling of epoxides with alkenylboronic acids

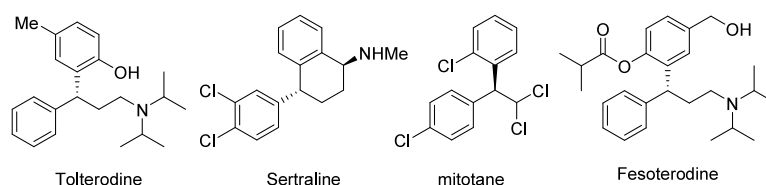


This work:

d. Ni-catalyzed asymmetric Suzuki-type-coupling of epoxides with arylboronic acids



e. Representative bioactive compounds containing chiral 1,1-diaryl alkane



Additionally, in 2019, the Zhang group developed a nickel-catalyzed stereoconvergent cross-coupling reaction between specific spiroepoxyoxindoles and alkenylboronic acids (Scheme 1c).¹¹ Despite advancements in the asymmetric ring-opening of epoxides, general protocols for the enantioselective coupling of racemic epoxides with arylboronic acids remain scarce. Therefore, the development of new transition-metal-catalyzed asymmetric Suzuki–Miyaura cross-coupling that addresses the

unmet need for versatile methodologies targeting epoxides to access chiral scaffolds is highly desirable and scientifically significant. Building on our previous work on nickel-catalyzed Suzuki-type cross-coupling of epoxides,¹² we hypothesized that incorporating a chiral nickel catalyst could establish a new paradigm for the enantioselective ring-opening of epoxides with arylboronic acids.

Table 1. Optimization of the Reaction Conditions^a

 L1 L2 L3 L4 L5 L6 L7 L8 L9 L10 L11 L12					
entry	catalyst	ligand	T (°C)	yield (%)	ee (%)
1	NiBr ₂ (diglyme)	L1	70	trace	
2	NiBr ₂ (diglyme)	L2	70	trace	
3	NiBr ₂ (diglyme)	L3	70	trace	
4	NiBr ₂ (diglyme)	L4	70	<3	
5	NiBr ₂ (diglyme)	L5	70	55	81
6	NiBr ₂ (diglyme)	L6	70	54	82
7	NiBr ₂ (diglyme)	L7	70	42	69
8	NiBr ₂ (diglyme)	L8	70	32	68
9	NiBr ₂ (diglyme)	L9	70	56	59
10	NiBr ₂ (diglyme)	L10	70	68	83
11	NiBr ₂ (diglyme)	L10	60	67	83
12	NiBr ₂ (diglyme)	L10	50	65	86
13	NiBr ₂ (diglyme)	L10	40	38	85
14	NiBr ₂ (diglyme)	L11	50	66	88
15	NiBr ₂ (diglyme)	L12	50	78	93
16 ^b	[Ni]	L12	50	76	93
17	NiCl ₂	L12	50	63	92
18	NiCl ₂ (DME)	L12	50	72	93
19	w/o	L12	50	0	
20	NiBr ₂ (diglyme)	w/o	50	0	

^aReaction conditions: **1a** (0.3 mmol, 1.5 equiv), **2a** (0.2 mmol, 1.0 equiv), catalyst (8 mol %), ligand (10 mol %), solvent (0.8 mL, 0.4 M). Isolated yield. ee value was determined by chiral HPLC. Reactions were carried out under an argon atmosphere. ^b[Ni] = Ni(ClO₄)₂·6H₂O.

Herein, we report a unique combination of an epoxide electrophile, an arylboronic acid nucleophile, and an exogenous base-free, enantioselective Ni-catalyzed Suzuki-type coupling reaction (Scheme 1d). This protocol enables the efficient construction of important chiral 1,1-diarylalkane scaffolds¹³ with high regioselectivity and excellent enantioselectivity (Scheme 1e). Mechanistic studies reveal that the reaction proceeds via iodide-mediated nucleophilic ring-opening of epoxides, followed by stereoconvergent radical cross-coupling

with arylboronic acids.^{10,14} Notably, this protocol operates under mild conditions and demonstrates broad functional group tolerance, making it applicable in the synthesis of pharmaceutical intermediates.

RESULTS AND DISCUSSION

To develop an efficient nickel-catalyzed enantioselective ring-opening of styrene oxides with arylboronic acids, we began our study by evaluating a range of chiral ligands using 4-

biphenylboronic acid (**1a**) and styrene oxide (**2a**) as model coupling partners, employing conditions derived from our work (Table 1). Initial screening experiments were conducted at 70 °C using NiBr₂(diglyme) as the catalyst, alongside a series of common chiral amine-derived bidentate and tridentate ligands. These studies revealed that both yield and enantioselectivity were significantly dependent on the choice of ligand. Evaluation of various ligand scaffolds demonstrated that conventional bis(oxazoline) ligands (BiOx and BOX, **L1** and **L2**), bis(oxazolyl)pyridyl ligand (PyBOX, **L3**), and pyridine-oxazoline-type ligand (PyrOx, **L4**) were ineffective for this enantioselective ring-opening reaction (entries 1–4). In contrast, the chiral biimidazoline (BiIm) ligand¹⁵ (entry 5) provided a moderate yield (**3a**, 55%) with favorable enantioselectivity (81% ee). Encouraged by these results, we screened additional chiral BiIm ligands (entries 6–10, **L6**–**L10**). Notably, **L10** delivered the highest yield (68%) with 83% ee, whereas ligands **L6**–**L9** yielded moderate results (32–56%) with suboptimal enantiomeric excesses (59–82% ee), indicating that the steric and electronic properties of the ligand significantly influence both activity and stereocontrol. We then explored the effect of temperature using **L10**, discovering that lowering the temperature to 50 °C further improved enantioselectivity to 86% ee (entry 12). However, reducing the temperature to 40 °C markedly decreased reactivity (entry 13, with a 38% yield). Subsequently, we tested additional BiIm ligands (**L11**–**L12**) at 50 °C (entries 14–15). Ligand **L11** provided a yield of 66% with 88% ee, while **L12** enhanced both metrics to 78% yield and 93% ee. To assess the influence of the nickel source, we screened various nickel precatalysts with **L12** at 50 °C (entries 16–18). Ni(ClO₄)₂·6H₂O and NiCl₂(dimethoxyethane) afforded high yields (76 and 72%, respectively) and excellent enantioselectivities (92–93% ee), whereas NiCl₂ alone yielded 63% with 92% ee. Control experiments confirmed the necessity of both the nickel catalyst and the chiral ligand, as no product was detected in their absence (entries 19–20). Through systematic optimization, the optimal conditions were identified as NiBr₂(diglyme) combined with ligand **L12** at 50 °C, delivering the target product in 78% yield and 93% ee.

After establishing the optimal reaction conditions, we systematically investigated the substrate scope of this asymmetric ring-opening reaction, focusing on the coupling of styrene oxides with various arylboronic acids (Table 2). A wide array of arylboronic acids containing distinct functional groups demonstrated compatibility: both electron-donating groups, such as phenyl (**3a**, **4a**, and **5a**), ether (**8a**), and silicon-containing group (**9a**), as well as electron-withdrawing groups like fluoro (**5a**) and trifluoromethyl (**7a**), reacted smoothly with styrene oxide to yield the corresponding chiral products in good yields and satisfactory enantioselectivities. Importantly, arylboronic acids featuring synthetically valuable functional groups, including thioether (**10a**), aldehyde (**11a**), ketone (**12a** and **13a**), ester (**14a**), amide (**17a** and **18a**), and amino groups (–NH₂, **19a**), were well-tolerated under the standard conditions, producing chiral 2,2-diaryl alcohols with yields ranging from 54 to 82% and enantiomeric excesses (ee) of 89–94%. Additionally, arylboronic acids with halogen substituents—including aryl chloride (**6a**), aryl bromide (**15a**), and aryl iodide (**16a**)—also participated efficiently in the reaction, yielding the corresponding β,β-diaryl alcohols in moderate to good yields and high enantioselectivity. The retention of halogen groups in the products provides

Table 2. Scope of the Enantioselective Suzuki–Miyaura Coupling of Arylboronic Acids^a

3a , 78%, 93% ee 42%, 90% ee ^b	4a , 80%, 92% ee	5a , 73%, 90% ee	6a , 74%, 91% ee
7a , 68%, 91% ee	8a , 69%, 92% ee	9a , 73%, 90% ee	10a , 70%, 92% ee
11a , 61%, 89% ee	12a , 66%, 92% ee	13a , 82%, 94% ee XRD single crystal diffraction CCDC# 2486619	14a , 68%, 93% ee
15a , 56%, 89% ee	16a , 45%, 88% ee	17a , 60%, 91% ee	18a , 54%, 90% ee
19a , 58%, 94% ee	20a , 67%, 92% ee	21a , 66%, 91% ee	22a , 55%, 90% ee
23a , 50%, 87% ee	24a , 71%, 90% ee	25a , 57%, 92% ee	26a , 36%, 87% ee

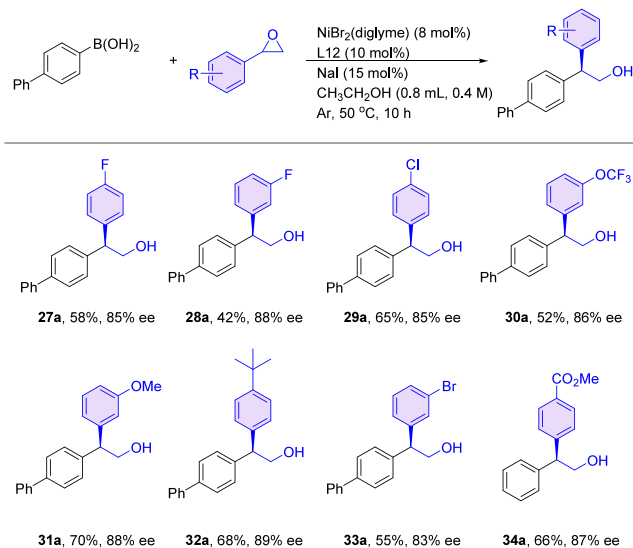
^aAll the reactions were carried out with arylboronic acids (0.3 mmol), epoxides (0.20 mmol), NiBr₂(diglyme) (0.016 mmol), **L12** (0.02 mmol), and NaI (0.03 mmol) in CH₃CH₂OH (0.8 mL) at 50 °C for 10 h. ^b4 mmol scale with 1 mol % catalyst loading.

opportunities for further late-stage derivatization via cross-coupling or other halogen-specific transformations. Heterocyclic arylboronic acids containing nitrogen, oxygen, or sulfur—such as benzofuran (**20a**), benzothiophene (**21a**), dibenzothiophene (**22a**), indole (**23a**), pyridine (**24a**), indazole (**25a**), and benzothiazole (**26a**)—also demonstrated compatibility, highlighting the protocol's potential for synthesizing bioactive molecules. Structural confirmation of the products revealed that the absolute configuration of compound **13a** was unambiguously determined by X-ray single-crystal diffraction, and the configurations of all other products were assigned by analogy to that of **13a**. However, vinylboronic acid substrates yielded products with only moderate enantioselectivity.¹⁰ Additionally, ortho-substituted arylboronic acids yielded only trace amounts of the target product, presumably

due to steric hindrance between the ortho substituent and the catalyst or epoxide substrate.

Next, we evaluated the substrate scope of styrene oxides, with results summarized in Table 3. A diverse set of styrene

Table 3. Scope of the Enantioselective Coupling of Epoxides^a



^aAll the reactions were carried out with arylboronic acids (0.3 mmol), epoxides (0.20 mmol), $\text{NiBr}_2(\text{diglyme})$ (0.016 mmol), **L12** (0.02 mmol), and NaI (0.03 mmol) in $\text{CH}_3\text{CH}_2\text{OH}$ (0.8 mL) at 50 °C for 10 h.

oxides bearing substituents such as fluoro (**27a** and **28a**), chloro (**29a**), bromo (**33a**), trifluoromethoxy (**30a**), methoxy (**31a**), *tert*-butyl (**32a**), and ester (**34a**) groups proved compatible with the optimized protocol. These substrates underwent smooth reactions, yielding the corresponding chiral products in moderate to good yields with satisfactory enantioselectivities. However, styrene oxides bearing ortho-substituents showed limited reactivity.

To further validate the synthetic utility of this asymmetric ring-opening cross-coupling, we extended the study to structurally complex substrates derived from natural products and pharmaceuticals (Scheme 2a). For example, an estrone-derived styrene oxide reacted efficiently with arylboronic acid under standard conditions, affording the target product (**35a**) in 56% isolated yield and excellent diastereoselectivity. Similarly, an indometacin-derived arylboronic acid was successfully incorporated into the reaction (**36a**), maintaining high stereocontrol. Furthermore, to demonstrate the efficiency and applicability of the present methodology, we conducted a scale-up synthesis of **37a**, affording the product in good synthetically useful yield alongside excellent enantiomeric excess. The product **37a** was successfully converted into **38a**, which could then be readily transformed into (+)-indatraline (a nonselective monoamine transporter inhibitor) and (+)-sertraline (a clinically important commercial drug for the treatment of depression, and substance dependency) following reported procedures (Scheme 2b).¹⁶

Beyond styrene oxides, styrenyl aziridines were also validated as viable coupling partners in this transformation. Specifically, several arylboronic acids reacted smoothly with styrenyl aziridines, affording the corresponding chiral products

in moderate yields and satisfactory enantioselectivities (Table 4). These results underscore the broader applicability of the developed catalytic system, enabling efficient construction of diverse chemical scaffolds bearing enantioenriched 1,1-diaryl motifs. Such versatility significantly enhances the synthetic utility of this protocol for accessing chiral building blocks with varied heteroatom-containing backbones.¹⁷

To elucidate the mechanism of the ring-opening reaction, a series of control experiments were conducted. First, enantiopure (R)-**2a** and (S)-**2a** were separately subjected to the optimized asymmetric reaction conditions.¹⁰ Notably, both enantiomeric substrates afforded the target product with comparable yield and enantioselectivity (Scheme 3a). Critically, the absolute configuration of the major product enantiomer was solely dictated by the chiral ligand, rather than the stereochemistry of the starting epoxides (**2a**). This observation confirms that the transformation proceeds via a stereoconvergent pathway, in which the chiral catalyst overrides the intrinsic stereochemistry of the racemic starting material to generate a single enantioenriched product.

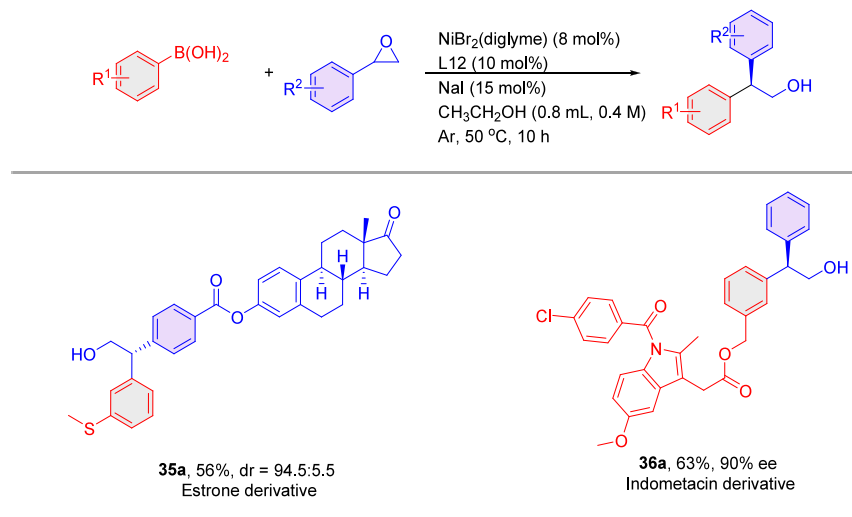
Additional control experiments were performed to probe alternative activation pathways for styrene oxides. When the reaction was conducted in the absence of NaI , no detectable product was formed, indicating that NaI plays an essential role in epoxide activation.^{12,18} Similarly, replacing the epoxide with presynthesized bromohydrin or iodohydrin intermediates as coupling partners failed to yield the target product; this outcome is tentatively attributed to the lack of a base required to drive the reaction forward. To validate this hypothesis, exogenous bases (e.g., K_2CO_3 , sodium ethoxide) were introduced. Among these additives, only sodium ethoxide enabled trace formation of the target product, affording it in 9% yield. We screened a series of bases, but the reactions only afforded very low yields (see SI). It is plausible that the reaction necessitates the in situ generation of a low-concentration iodohydrin intermediate. In addition, we detected the iodohydrin intermediate by high-resolution mass spectrometry (HRMS) (Scheme 3b).

To verify the involvement of radical intermediates, radical quenching experiments were conducted using 2,2,6,6-tetramethylpiperidinoxy (TEMPO), a well-characterized radical scavenger. Addition of TEMPO effectively inhibited the reaction, and a TEMPO-trapped adduct (**38a**) was further detected by HRMS, as shown in Scheme 3c. Furthermore, upon adding 5,5-dimethyl-1-pyrroline *N*-oxide (DMPO) to the model reaction, electron paramagnetic resonance (EPR) analysis of the mixture revealed a distinct radical signal. Based on the simulated spectrum, this signal was assigned to the trapped radical-DMPO adduct (Scheme 3d, see SI for details).¹⁹ Collectively, these results support two key mechanistic conclusions: (1) the reaction proceeds through a halohydrin intermediate, and (2) the base required for the transformation is derived from in situ-generated alkoxide anions formed during epoxide activation and ring-opening.

Nonlinear effect analysis was conducted to probe the nature of the catalytically active species, revealing a linear correlation between the enantiopurity of product **3a** and that of the chiral ligand.^{17,20} Additionally, the reaction's enantioselectivity remained unchanged even when ligand loading was varied, including at stoichiometric excess of the ligand (Scheme 3e). Collectively, these observations indicate that the nickel center coordinates with a single chiral ligand in the active catalytic species. To further validate the reaction pathway, the

Scheme 2. Synthetic Applications: (a) Late-Stage Functionalization; (b) Derivation of Products to the Pharmaceutical Intermediates

a. Late-stage functionalization



b. Derivation of products to the pharmaceutical intermediates

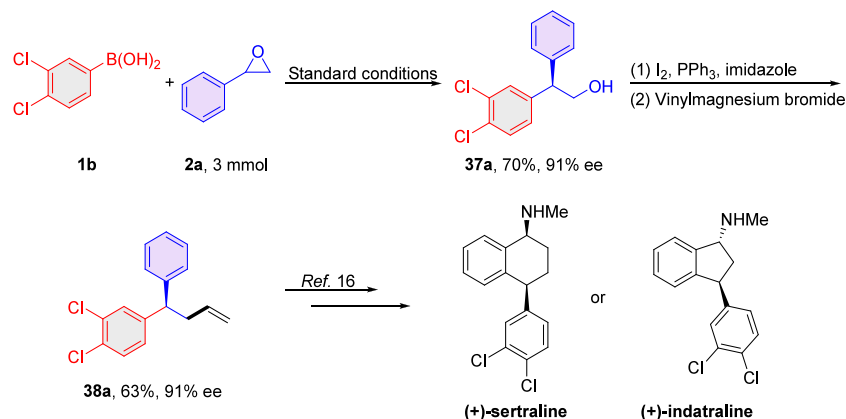
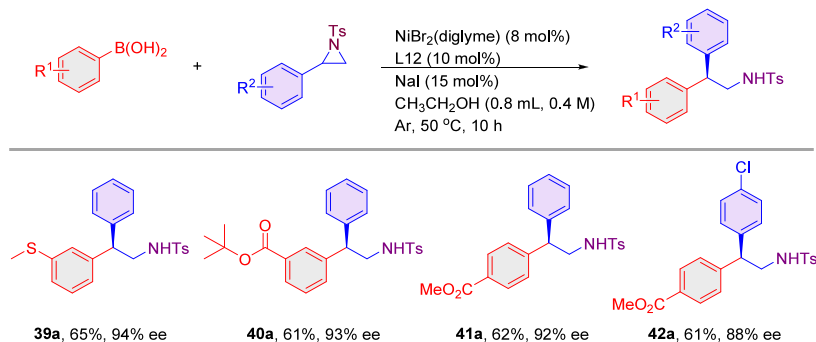


Table 4. Scope of the Enantioselective Coupling of Arylaziridines

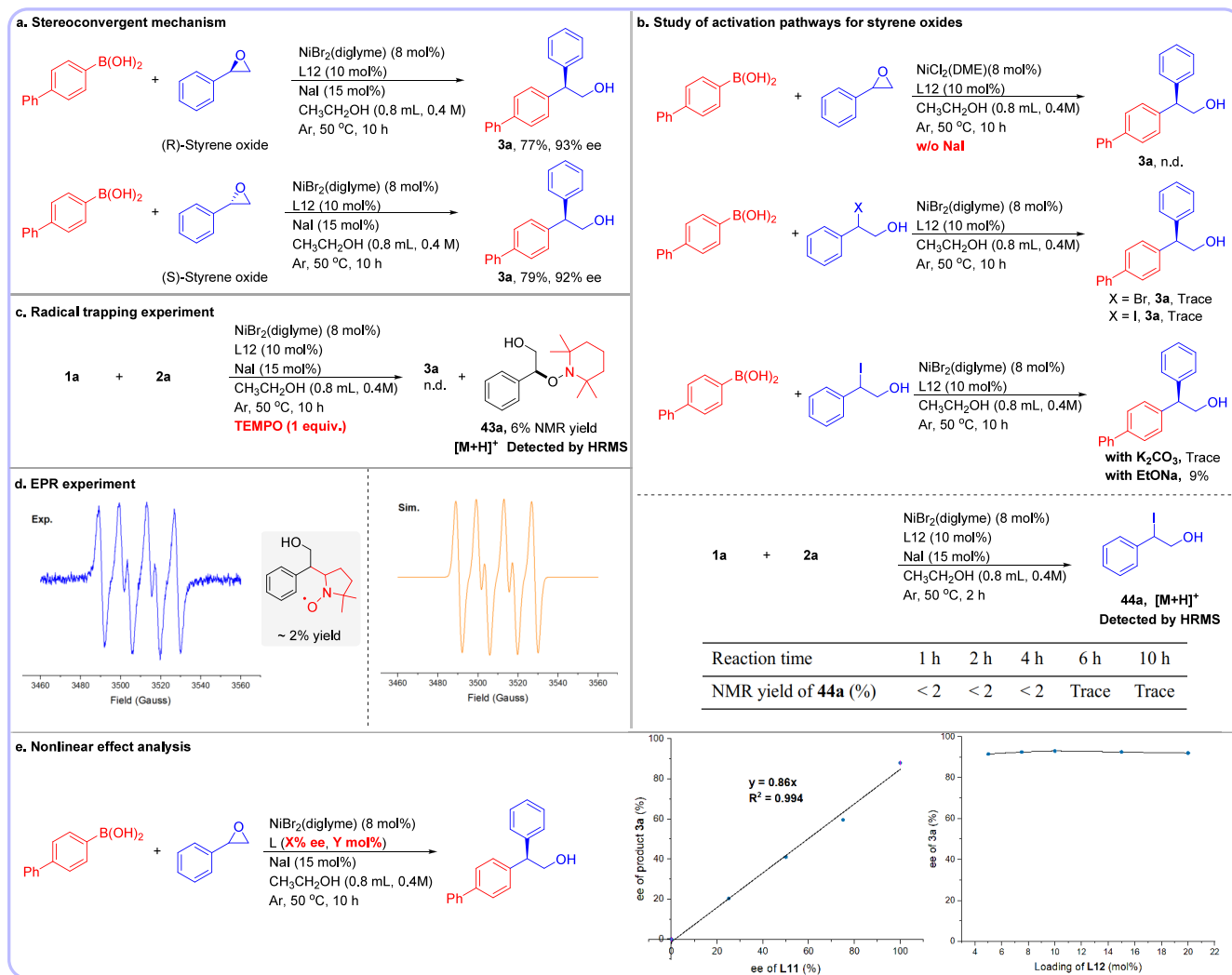


enantiomeric excess (ee) of product **3a** was monitored at various conversion stages. Throughout the reaction, the ee of **3a** remained consistent. This observation confirms that the transformation proceeds via an enantioconvergent pathway, with detailed data provided in the Supporting Information.

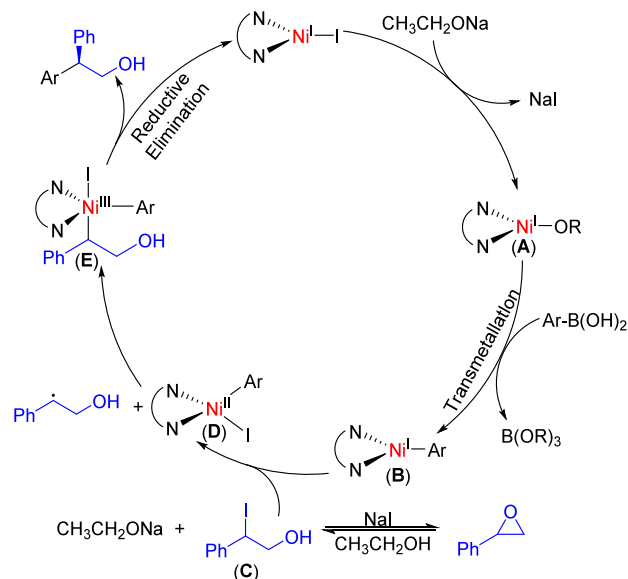
Based on the aforementioned experimental results and relevant literature reports,²¹ a plausible mechanism for this reaction is proposed in Scheme 4. The arylboronic acid undergoes transmetalation with an alkoxy nickel(I) species (A)

to generate an arylnickel(I) intermediate (B). Concurrently, the styrene oxide undergoes ring opening promoted by sodium iodide and ethanol to form an iodohydrin intermediate (C) and sodium alkoxide. This iodohydrin then undergoes oxidative addition with the arylnickel(I) catalyst, producing an alkyl radical and a nickel(II) species (D). The alkyl radical subsequently oxidizes the nickel(II) species to yield an organonickel(III) intermediate (E); reductive elimination from this organonickel(III) intermediate affords the target

Scheme 3. Mechanistic Studies: (a) Stereoconvergent Mechanism; (b) Study of Activation Pathways for Styrene Oxides; (c) Radical Trapping Experiment; (d) EPR Experiment; (e) Nonlinear Effect Analysis



Scheme 4. Proposed Reaction Mechanism



product. Additionally, an alternative mechanism initiated by oxidative addition to a nickel(0) catalyst cannot be excluded.

■ CONCLUSIONS

In summary, we have developed a practical nickel-catalyzed enantioselective Suzuki–Miyaura-type coupling of aryl epoxides with arylboronic acids, representing a significant advance in asymmetric C–C bond-forming reactions of epoxides. This transformation exhibits remarkable functional group tolerance and provides an efficient route to enantioenriched 2,2-diaryl alcohols and amines, which are key intermediates in the synthesis of pharmaceuticals, agrochemicals, and natural products. Mechanistic investigations provide insights into the reaction pathway, which proceeds via a stereoconvergent mechanism, involves halohydrin intermediates, and proceeds through radical intermediates. Notably, this reaction relies on in situ-generated alkoxide anions as the essential base for driving the reaction forward.

■ MATERIALS AND METHODS

See the Supporting Information for complete details, including all compound characterization and analytical data.

Preparation of Epoxides

Method A. A solution of NaH (11 mmol, 2.2 equiv, 60% dispersion in mineral oil) and Me₃Si (11 mmol, 2.2 equiv) in DMSO (7 mL) and THF (6 mL) was stirred for 30 min at room temperature in a 50 mL flask under an Ar atmosphere. Then a solution of aldehyde (5 mmol, 1.0 equiv) in THF was added dropwise at 0 °C. After complete addition, the mixture was warmed to room temperature and allowed to stir for 12 h. The mixture was quenched with sat. NH₄Cl and diluted with ethyl acetate. Then the mixture was separated and the aqueous layer was back-extracted with ethyl acetate. The organic layer was washed with brine and dried over Na₂SO₄. The solvent was removed under reduced pressure and the residue was purified by column chromatography to give the desired epoxides **1a–1h**.¹⁰

Method B. A 100 mL round-bottom flask, flame-dried and equipped with a stir bar, was charged with styrene (5 mmol, 1 equiv) dissolved in dichloromethane (30 mL). *m*-chloroperbenzoic acid (10 mmol, 2.0 equiv) was added, and the mixture was stirred at room temperature overnight. The reaction was quenched with saturated aqueous NaHCO₃ and extracted three times with dichloromethane. The combined organic layers were washed successively with water, saturated aqueous NaHCO₃, and brine, then dried over Na₂SO₄, filtered, and concentrated under reduced pressure. The residue was purified by column chromatography to afford the desired epoxides **1l**.¹⁰

Preparation of Tosylaziridine

A round-bottom flask was flame-dried, and the alkene (1.0 equiv), chloramine-T trihydrate (1.1 equiv), and acetonitrile (5 mL per mmol of alkene) were added. Trimethylphenylammonium tribromide (0.1 equiv) was then introduced, and the mixture was stirred at room temperature under argon overnight. The resulting mixture was filtered through a pad of Celite, and the filtrate was concentrated. The crude product was purified by column chromatography.¹⁷

Preparation of Chiral Ligand

L-(+)-Isuleucinol S1 (20 mmol, 2.34 g, 2.0 equiv) and dimethyloxalate (10 mmol, 1.18 g, 1.0 equiv) were dissolved in toluene (70 mL) and heated to 85 °C. The reaction was allowed to stir overnight and then cooled to room temperature with the diamide precipitating out of solution as a white solid, then filtered and washed with petroleum ether to get the diol S2 (2.73 g, 95% yield, white solid).

The diol S2 (10 mmol, 1.0 equiv) was dissolved in toluene (50 mL) and heated to 90 °C where upon thionyl chloride (1.72 mL, 2.4 equiv) was added. The reaction was stirred for 8 h. After the reaction is completed, it is cooled to room temperature, a large amount of petroleum ether is added, and then the dichloro intermediate S3 is precipitated and filtered to afford (2.91 g, 90% yield, white solid).

To an oven-dried flask was added S3 (6.0 mmol, 1.94 g, 1.0 equiv) and phosphorus pentachloride (14.4 mmol, 2.99 g, 2.4 equiv) in 30 mL of toluene under Ar atmosphere. The reaction was allowed to stir at 85 °C for 8 h before it was cooled to room temperature. Toluene was evaporated under reduced pressure (80% yield, yellow liquid).

To an oven-dried flask was added S4 (5 mmol), Et₃N (50 mmol, 10.0 equiv) and 3,5-ditert-butylaniline (20 mmol, 4.1 g, 4.0 equiv) in 25 mL of MeCN under Ar atmosphere. Next, the reaction mixture was heated to reflux under Ar for 24 h. After been cooled to ambient temperature, the volatiles were removed by rotary evaporation. The residue was diluted with CH₂Cl₂ washed with saturated NH₄Cl solution (100 mL × 3) and water. The organic layer was dried with Na₂SO₄, filtered and concentrated under reduced pressure. The residue was purified by flash column chromatography (petroleum ether: EtOAc = 3:1) and then recrystallized from hexane and EtOAc to give ligand **L12**.¹⁵

General Procedure for Enantioselective Coupling

In the air, a 10 mL Schlenk tube equipped with a magnetic stirrer was charged with NiBr₂(diglyme) (5.6 mg, 0.016 mmol, 8 mol %), L12 (12.5 mg, 0.02 mmol, 10 mol %), arylboronic acid (0.30 mmol, 1.50 equiv) and sodium iodide (4.5 mg, 0.03 mmol, 15 mol %) (Note a: If the epoxide was solid, it was also added at this step). The Schlenk tube was evacuated and backfilled with argon three times, then

ethanol (0.8 mL, 0.4 M) was added, followed by the epoxide (0.2 mmol, 1.0 equiv). Then the solution was stirred at 50 °C for 10 h. The resulting crude reaction was purified by column chromatography using ethyl acetate and petroleum ether as eluent, and analyzed by HPLC using a chiral stationary phase.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its online Supporting Information.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.5c07500>.

Details of experimental procedures and full characterization data and copies of NMR spectra; general information; reaction optimization; mechanistic studies; NMR and HPLC spectra; and crystallographic information file for **13a** (PDF)

Accession Codes

CCDC 2486619 contains the crystallographic data for this paper.

■ AUTHOR INFORMATION

Corresponding Author

Xiao-Yu Lu – School of Materials and Chemical Engineering, ChuZhou University, Chu Zhou 239000, China; orcid.org/0000-0001-6069-9739; Email: xiaoyulu@mail.ustc.edu.cn

Authors

Hao-Yu Dong – School of Materials and Chemical Engineering, ChuZhou University, Chu Zhou 239000, China
Jun-Long Zhang – School of Materials and Chemical Engineering, ChuZhou University, Chu Zhou 239000, China
Jin-Tao Tao – Institutes of Physical Science and Information Technology, Anhui University, Hefei, Anhui 230601, China
Si-Nan Liu – School of Materials and Chemical Engineering, ChuZhou University, Chu Zhou 239000, China
Yi-Mei Wu – Institutes of Physical Science and Information Technology, Anhui University, Hefei, Anhui 230601, China

Complete contact information is available at: <https://pubs.acs.org/10.1021/acscatal.5c07500>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We gratefully acknowledge financial support from the Excellent Youth Project of University Natural Science Research of Anhui Province (2023AH030095) and the National Natural Science Foundation of China (22001029).

■ REFERENCES

- (1) (a) Malapit, C. A.; Bour, J. R.; Brigham, C. E.; Sanford, M. S. Base-free nickel-catalysed decarbonylative Suzuki–Miyaura coupling of acid fluorides. *Nature* **2018**, 563, 100–104. (b) Rowsell, B. J. S.; O'Brien, H. M.; Athavan, G.; Daley-Dee, P. R.; Krieger, J.; Richards, E.; Heaton, K.; Fairlamb, I. J. S.; Bedford, R. B. The iron-catalysed Suzuki coupling of aryl chlorides. *Nat. Catal.* **2024**, 7, 1186–1198. (c) Cook, A.; St. Onge, P.; Newman, S. G. Deoxygenative Suzuki–Miyaura arylation of tertiary alcohols through silyl ethers. *Nat. Synth.*

- 2023, 2, 663–669. (d) Jana, R.; Pathak, T. P.; Sigman, M. S. Advances in Transition Metal (Pd,Ni,Fe)-Catalyzed Cross-Coupling Reactions Using Alkyl-organometallics as Reaction Partners. *Chem. Rev.* **2011**, 111, 1417–1492. (e) Suzuki, A. Cross-Coupling Reactions of Organoboranes: An Easy Way To Construct C–C Bonds (Nobel Lecture). *Angew. Chem., Int. Ed.* **2011**, 50, 6722–6737. (f) Lennox, A. J. J.; Lloyd-Jones, G. C. Transmetalation in the Suzuki–Miyaura Coupling: The Fork in the Trail. *Angew. Chem., Int. Ed.* **2013**, 52, 7362–7370. (g) Han, F.-S. Transition-metal-catalyzed Suzuki–Miyaura cross-coupling reactions: a remarkable advance from palladium to nickel catalysts. *Chem. Soc. Rev.* **2013**, 42, 5270–5298. (h) Meringdal, J. W.; Menche, D. Suzuki–Miyaura (hetero-)aryl cross-coupling: recent findings and recommendations. *Chem. Soc. Rev.* **2025**, 54, 5746–5765.
- (2) (a) Wang, Y.; Haight, I.; Gupta, R.; Vasudevan, A. What is in Our Kit? An Analysis of Building Blocks Used in Medicinal Chemistry Parallel Libraries. *J. Med. Chem.* **2021**, 64, 17115–17122. (b) Brown, D. G.; Boström, J. Analysis of Past and Present Synthetic Methodologies on Medicinal Chemistry: Where Have All the New Reactions Gone? *J. Med. Chem.* **2016**, 59, 4443–4458. (c) Roughley, S. D.; Jordan, A. M. The Medicinal Chemist's Toolbox: An Analysis of Reactions Used in the Pursuit of Drug Candidates. *J. Med. Chem.* **2011**, 54, 3451–3479. (d) Cooper, T. W. J.; Campbell, I. B.; Macdonald, S. J. F. Factors Determining the Selection of Organic Reactions by Medicinal Chemists and the Use of These Reactions in Arrays (Small Focused Libraries). *Angew. Chem., Int. Ed.* **2010**, 49, 8082–8091.
- (3) (a) Gorzynski Smith, J. Synthetically Useful Reactions of Epoxides. *Synthesis* **1984**, 1984, 629–656. (b) Huang, C.-Y.; Doyle, A. G. The Chemistry of Transition Metals with Three-Membered Ring Heterocycles. *Chem. Rev.* **2014**, 114, 8153–8198. (c) Crotti, P.; Pineschi, M. Epoxides in Complex Molecule Synthesis. *Aziridines and Epoxides in Organic Synthesis* **2006**, 271–313. (d) Nielsen, D. K.; Doyle, A. G. Nickel-Catalyzed Cross-Coupling of Styrenyl Epoxides with Boronic Acids. *Angew. Chem., Int. Ed.* **2011**, 50, 6056–6059.
- (4) (a) Jacobsen, E. N. Asymmetric Catalysis of Epoxide Ring-Opening Reactions. *Acc. Chem. Res.* **2000**, 33, 421–431. (b) Lidskog, A.; Li, Y.; Wärnmark, K. Asymmetric Ring-Opening of Epoxides Catalyzed by Metal–Salen Complexes. *Catalysts* **2020**, 10, 705. (c) Zhu, Y.; Wang, Q.; Cornwall, R. G.; Shi, Y. Organocatalytic Asymmetric Epoxidation and Aziridination of Olefins and Their Synthetic Applications. *Chem. Rev.* **2014**, 114, 8199–8256. (d) Nielsen, L. P. C.; Jacobsen, E. N. Catalytic Asymmetric Epoxide Ring-opening Chemistry. In *Aziridines and Epoxides in Organic Synthesis* **2006**, 229–269.
- (5) (a) Pastor, I. M.; Yus, M. Asymmetric Ring Opening of Epoxides. *Curr. Org. Chem.* **2005**, 9, 1–29. (b) Teng, F.; Li, Z.-Q.; Mao, Z.-C.; Zhang, Z.; Jiang, M.; Xiao, W.-J.; Chen, J.-R. Unlocking Epoxide Radical Anion Reactivity for Asymmetric Cross-Coupling by Copper Metallaphotoredox. *J. Am. Chem. Soc.* **2025**, 147, 19400–19412.
- (6) (a) Dilek, Ö.; Tezeren, M. A.; Tilki, T.; Ertürk, E. Chiral 2-(2-hydroxyaryl)alcohols (HAROLs) with a 1,4-diol scaffold as a new family of ligands and organocatalysts. *Tetrahedron* **2018**, 74, 268–286. (b) Oguni, N.; Miyagi, Y.; Itoh, K. Highly enantioselective arylation of symmetrical epoxides with phenyllithium promoted by chiral schiff bases and salens. *Tetrahedron Lett.* **1998**, 39, 9023–9026.
- (7) (a) Zhang, L.; Wang, W.; Shen, C.; Dong, K. Enantioconvergent Cross-Electrophile Coupling of 2-Aryloxetanes with Aryl and Vinyl Halides, or Anhydrides. *ACS Catal.* **2025**, 15, 7578–7587. (b) Shen, J.; Xu, Z.; Yang, S.; Li, S.; Jiang, J.; Zhang, Y.-Q. Quaternary Stereocenters via Catalytic Enantioconvergent Allylation of Epoxides. *J. Am. Chem. Soc.* **2023**, 145, 21122–21131. (c) Li, J.; Li, S.; Cui, G.-H.; Yuan, K.; Wang, C.; Yang, C.; Zhang, G.; Guo, R. Photoinduced Copper-Catalyzed Regio- and Enantioselective Alkynylation of Epoxides. *ACS Catal.* **2024**, 14, 7553–7561.
- (8) Zhao, Y.; Weix, D. J. Enantioselective Cross-Coupling of meso-Epoxides with Aryl Halides. *J. Am. Chem. Soc.* **2015**, 137, 3237–3240.
- (9) Banerjee, A.; Yamamoto, H. Nickel Catalyzed Regio-, Diastereo-, and Enantioselective Cross-Coupling of 3,4-Epoxyalcohol with Aryl Iodides. *Org. Lett.* **2017**, 19, 4363–4366.
- (10) Lau, S. H.; Borden, M. A.; Steiman, T. J.; Wang, L. S.; Parasram, M.; Doyle, A. G. Ni/Photoredox-Catalyzed Enantioselective Cross-Electrophile Coupling of Styrene Oxides with Aryl Iodides. *J. Am. Chem. Soc.* **2021**, 143, 15873–15881.
- (11) Wu, L.; Yang, G.; Zhang, W. Ni-Catalyzed Enantioconvergent Coupling of Epoxides with Alkenylboronic Acids: Construction of Oxindoles Bearing Quaternary Carbons. *CCS Chem.* **2020**, 2, 623–631.
- (12) Lu, X.-Y.; Yan, L.-Y.; Li, J.-S.; Li, J.-M.; Zhou, H.-p.; Jiang, R.-C.; Liu, C.-C.; Lu, R.; Hu, R. Base-free Ni-catalyzed Suzuki-type cross-coupling reactions of epoxides with boronic acids. *Chem. Commun.* **2020**, 56, 109–112.
- (13) (a) Cheng, X.; Lu, H.; Lu, Z. Enantioselective benzylic C–H arylation via photoredox and nickel dual catalysis. *Nat. Commun.* **2019**, 10, 3549. (b) Poremba, K. E.; Kadunce, N. T.; Suzuki, N.; Cherney, A. H.; Reisman, S. E. Nickel-Catalyzed Asymmetric Reductive Cross-Coupling to Access 1,1-Diaryllalkanes. *J. Am. Chem. Soc.* **2017**, 139, 5684–5687. (c) Gu, X.-H.; Yu, H.; Jacobson, A. E.; Rothman, R. B.; Dersch, C. M.; George, C.; Flippen-Anderson, J. L.; Rice, K. C. Design, Synthesis, and Monoamine Transporter Binding Site Affinities of Methoxy Derivatives of Indatraline. *J. Med. Chem.* **2000**, 43, 4868–4876. (d) Abad, A.; López-Pérez, J. L.; del Olmo, E.; García-Fernández, L. F.; Francesch, A.; Trigili, C.; Barasoain, I.; Andreu, J. M.; Díaz, J. F.; San Feliciano, A. Synthesis and Antimitotic and Tubulin Interaction Profiles of Novel Pinacol Derivatives of Podophyllotoxins. *J. Med. Chem.* **2012**, 55, 6724–6737. (e) Silva, D. H. S.; Davino, S. C.; Barros, S. B. d. M.; Yoshida, M. Dihydrochalcones and Flavonolignans from *Iryanthera lancifolia*. *J. Nat. Prod.* **1999**, 62, 1475–1478. (f) Ameen, D.; Snape, T. J. Chiral 1,1-diaryl compounds as important pharmacophores. *MedChemComm* **2013**, 4, 893–907.
- (14) (a) Hu, X.; Cheng-Sánchez, I.; Cuesta-Galisteo, S.; Nevado, C. Nickel-Catalyzed Enantioselective Electrochemical Reductive Cross-Coupling of Aryl Aziridines with Alkenyl Bromides. *J. Am. Chem. Soc.* **2023**, 145, 6270–6279. (b) Lan, Y.; Han, Q.; Liao, P.; Chen, R.; Fan, F.; Zhao, X.; Liu, W. Nickel-Catalyzed Enantioselective C(sp³)–C(sp³) Cross-Electrophile Coupling of N-Sulfonyl Styrenyl Aziridines with Alkyl Bromides. *J. Am. Chem. Soc.* **2024**, 146, 25426–25432.
- (15) (a) Li, T.; Luo, L.; Cheng, X.; Lu, Z. Enantioconvergent Cross-Coupling Reaction with 1,4-Dihydropyridine Derivatives via Photoinduced Nickel Catalysis. *ACS Catal.* **2024**, 14, 3278–3286. (b) Li, J.; Yu, B.; Lu, Z. Chiral Imidazoline Ligands and Their Applications in Metal-Catalyzed Asymmetric Synthesis. *Chin. J. Chem.* **2021**, 39, 488–514. (c) Boland, N. A.; Casey, M.; Hynes, S. J.; Matthews, J. W.; Müller-Bunz, H.; Wilkes, P. Preparation of enantiopure biimidazoline ligands and their use in asymmetric catalysis. *Org. Biomol. Chem.* **2004**, 2, 1995–2002. (d) Hao, X.-Q.; Dong, Y.-N.; Gao, B.; Li, K.; Zhao, X.-M.; Xu, Y.; Song, M.-P. Biimidazoline ligands for palladium-catalyzed asymmetric allylic alkylation. *Tetrahedron: Asymmetry* **2015**, 26, 1360–1368.
- (16) (a) Roesner, S.; Casatejada, J. M.; Elford, T. G.; Sonawane, R. P.; Aggarwal, V. K. Enantioselective Syntheses of (+)-Sertraline and (+)-Indatraline Using Lithiation/Borylation–Protodeboration Methodology. *Org. Lett.* **2011**, 13, 5740–5743. (b) Froimowitz, M.; Wu, K.-M.; Moussa, A.; Haidar, R. M.; Jurayj, J.; George, C.; Gardner, E. L. Slow-Onset, Long-Duration 3-(3',4'-Dichlorophenyl)-1-indanamine Monoamine Reuptake Blockers as Potential Medications to Treat Cocaine Abuse. *J. Med. Chem.* **2000**, 43, 4981–4992.
- (17) (a) Wang, Y.-Z.; Wang, Z.-H.; Eshel, I. L.; Sun, B.; Liu, D.; Gu, Y.-C.; Milo, A.; Mei, T.-S. Nickel/biimidazole-catalyzed electrochemical enantioselective reductive cross-coupling of aryl aziridines with aryl iodides. *Nat. Commun.* **2023**, 14, 2322. (b) Zhang, L.; Wang, H.; Santiago, T. G.; Yue, W.-J.; Martin, R. Photoinduced nickel-catalysed enantioconvergent sp³–sp³ coupling of unactivated olefins and aziridines. *Nat. Catal.* **2025**, 8, 348–356. (c) Ren, X.; Qiao, Y.; Zhou, B.; Liu, S.; Huang, Y. Nickel-catalysed enantioselective

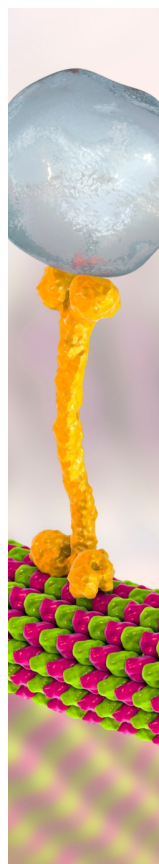
migratory reductive cross-coupling enabled by radical 1,2-amino migration. *Nat. Synth.* **2025**, *4*, 1232–1246. (d) Takeda, Y.; Ikeda, Y.; Kuroda, A.; Tanaka, S.; Minakata, S. Pd/NHC-Catalyzed Enantio-specific and Regioselective Suzuki–Miyaura Arylation of 2-Arylaziridines: Synthesis of Enantioenriched 2-Arylphenethylamine Derivatives. *J. Am. Chem. Soc.* **2014**, *136*, 8544–8547. (e) Woods, B. P.; Orlandi, M.; Huang, C.-Y.; Sigman, M. S.; Doyle, A. G. Nickel-Catalyzed Enantioselective Reductive Cross-Coupling of Styrenyl Aziridines. *J. Am. Chem. Soc.* **2017**, *139*, 5688–5691.

(18) (a) Lu, X.-Y.; Li, J.-S.; Wang, S.-Q.; Zhu, Y.-J.; Li, Y.-M.; Yan, L.-Y.; Li, J.-M.; Wang, J.-Y.; Zhou, H.-P.; Ge, X.-T. Pd-Catalyzed decarboxylative cross-coupling reactions of epoxides with α,β -unsaturated carboxylic acids. *Chem. Commun.* **2019**, *55*, 11123–11126. (b) Lu, X.-Y.; Shui, F.-Y.; Zhang, X.; Huang, R.; Wang, Z.-Z.; Song, Y.-W.; Jiang, F.; Yang, G.-X.; Sun, Y.-X. Palladium catalyzed tandem cyclization of acryloylbenzamides and N-arylacrylamides with epoxides: access to functionalized isoquinolinediones and oxindoles. *Org. Chem. Front.* **2023**, *10*, 5616–5622.

(19) (a) Wang, Q.; Choi, H.; Lyu, X.; Kim, D.; Chang, S. NiH-Catalyzed Homobenzylic Hydroalkylation of Aryl Alkenes Using Sulfoxonium Ylides. *J. Am. Chem. Soc.* **2025**, *147*, 38817–38826. (b) Cao, B.; Tan, H.-D.; Liu, G.; Leng, X.; Xue, X.-S.; Huang, Z. An ECEC Mechanism via an Electrophilic Co(II) Carbene in Reductive Cyclopropanation. *J. Am. Chem. Soc.* **2025**, *147*, 43640–43654. (c) Thai, P.; Frey, B. L.; Lalis, R. F.; Patel, L.; Mukherjee, P.; Song, Z.; Carmeli, R.; Gutierrez, O.; Powers, D. C. Multisite Proton-Coupled Electron Transfer Enables Iodanyl Radical Catalysis. *J. Am. Chem. Soc.* **2025**, *147*, 44336–44345.

(20) (a) Li, L.; Hu, Z.; Ren, S.; Arndtsen, B. A.; Chu, L. Enantioselective Carbonylative Coupling Reactions: Merging Nickel-Based Selectivity and Photoredox Reactivity. *J. Am. Chem. Soc.* **2025**, *147*, 28206–28214. (b) Fang, J.-D.; Pang, X.; Shu, X.-Z. Nickel-Catalyzed Enantioconvergent Cross-Electrophile Coupling of Benzylic Alcohols with Alkenyl Electrophiles. *J. Am. Chem. Soc.* **2025**, *147*, 28313–28321.

(21) (a) Mills, L. R.; Simmons, E. M.; Lee, H.; Nester, E.; Kim, J.; Wisniewski, S. R.; Pecoraro, M. V.; Chirik, P. J. (Phenoxyimine)-nickel-Catalyzed $C(sp^2)$ – $C(sp^3)$ Suzuki–Miyaura Cross-Coupling: Evidence for a Recovering Radical Chain Mechanism. *J. Am. Chem. Soc.* **2024**, *146*, 10124–10141. (b) Wilsily, A.; Tramutola, F.; Owston, N. A.; Fu, G. C. New Directing Groups for Metal-Catalyzed Asymmetric Carbon–Carbon Bond-Forming Processes: Stereoconvergent Alkyl–Alkyl Suzuki Cross-Couplings of Unactivated Electrophiles. *J. Am. Chem. Soc.* **2012**, *134*, 5794–5797. (c) Liu, D.; Wang, J.-W.; Lu, X.; Fu, Y. Entropy effects in temperature-regulated nickel-catalyzed regiodivergent alkene hydroalkylation. *Org. Chem. Front.* **2024**, *11*, 3648–3654. (d) Zhou, Y.; Qiu, L.; Li, J.; Xie, W. A General Copper Catalytic System for Suzuki–Miyaura Cross-Coupling of Unactivated Secondary and Primary Alkyl Halides with Arylborons. *J. Am. Chem. Soc.* **2023**, *145*, 28146–28155. (e) Wong, C. D.; Bradford, L. C.; Hirbawi, N.; Jarvo, E. R. Nickel-Catalyzed Suzuki–Miyaura Cross-Coupling Reaction of Aliphatic Alcohol Derivatives. *Angew. Chem., Int. Ed.* **2025**, *64*, No. e202509657. (f) Gao, G.; Li, Z.; Li, Y.; Lu, X. Recent Progress on the Construction of $C(sp^2)$ – $C(sp^3)$ Bonds via Photoredox/Nickel Dual Catalysis. *Chin. J. Org. Chem.* **2025**, *45*, 1905–1919.



CAS BIOFINDER DISCOVERY PLATFORM™

BRIDGE BIOLOGY AND CHEMISTRY FOR FASTER ANSWERS

Analyze target relationships,
compound effects, and disease
pathways

Explore the platform

CAS
A Division of the
American Chemical Society